

Free hosting

Many online services such as BitBucket and Google Code, provide free hosting for open source projects

and establish a time line. Access restrictions are also easier to establish, as the server is in control of a single entity.

However, this is also akin to putting all your eggs in to one basket, as all the content and its revision history is maintained at one single place.

One of the most popular examples of this is Subversion or SVN.

Decentralised revisioning

In the essence of P2P, distributed revision control systems give people equal power. It is in essence a true democracy. These are much more recent; however, they are seeing rapid adoption in the development community.

In the distributed system, as the name suggests, there is no need for a central repository, although one may be present. Instead, individuals maintain their own copy of the repository where they can make changes without the need for a network connection.

With decentralisation, changes can not only be made to and loaded from a repository, but also between repositories. Using the “push” operation, the changes made in one repository (which could consist of multiple commits) can be transmitted to another repository, and using the “pull” operation, changes can similarly be merged with your own repository.

This may seem like a nightmare if you imagine even a small group of 10 people pushing and pulling each other and committing, syncing, merging and branching. With such a cacophony of actions it seems more like a soap opera than a team programming effort!

To simplify this you can of course have a central repository for storing the final work of the team, pretty much the same as a central system.

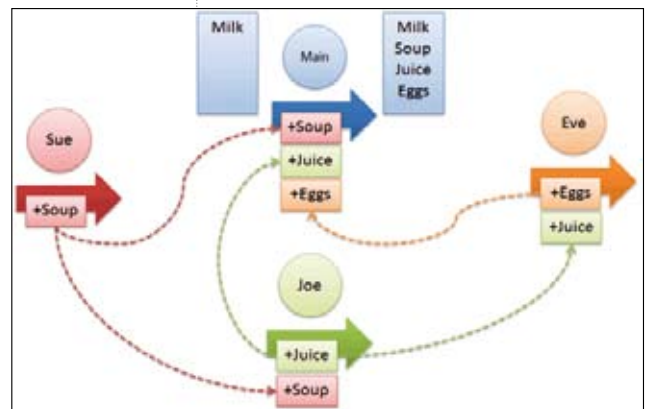
GIT is a distributed revision control system designed by Linus Torvalds, the creator of Linux, which is used for maintaining the Linux kernel. Another popular example of a distributed system is Mercurial, now used for Mozilla projects.

Evidently, distributed systems are perfect for single user scenarios. You can easily set up such a system and use it to maintain, for example, your documents folder.

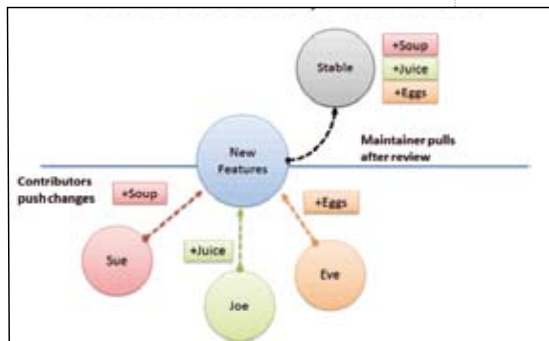
Conclusion

As you begin work on your next masterpiece keep in mind, how revision control software can play a vital role. These systems are usually associated only with large software projects, but you can clearly see how they can be useful even to the most casual computer users.

Even without being able to employ it to the fullest, and with little need for branching and tagging in a casual or individual work environment, the advantages are enough to justify the learning curve. Don't dismiss these systems as something that can only be of advantage to developers, find your own usage scenarios. Think of it as a supercharged backup solution (although it is not to be taken as a replacement for one!). May your future come in multiple revisions.

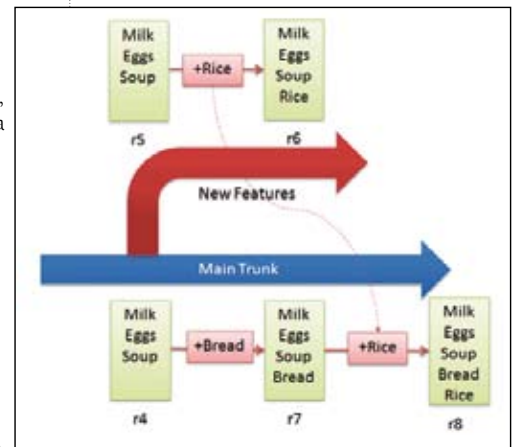


The distributed system



Distributed push and pull

Each person is in charge of their own repositories which they maintain. However, you may wonder, how does work get done in a group if each person works independently.



Merging changes

Mercurial

Mercurial is a decentralised system, despite the apparent complexity, it is a much simpler and faster system to set up and use.

The simplest way to use Mercurial, and one which will appease GUI buffs as well, is by using the TortoiseHG tool. This software not only installs the Mercurial system, but also installs extensions to Windows Explorer. As a result, it makes it much simpler to perform the most common tasks. Using TortoiseHG you can manage your repositories easily from explorer.

After installing TortoiseHG, all you need to do to enable revision control is to simply right click in a folder and under the TortoiseHG menu click on “Create Repository Here”. That's it! After a repository is created in a folder, you can instruct it to add the files to the repository using the Commit command.

Mercurial comes out as a much simpler solution for single people maintaining a project, however as you bring in multiple people, each with their own version of the repository, SVN may seem worth the whole effort of setting up a server.

Collaboration in Google Wave

Google Wave is a powerful service indeed, and being open source and free, it has the potential to change the way the internet works. It is powerful enough to replace current email and IM systems and perhaps change a lot more.

Google Wave is a powerful collaboration tool, where teamwork and versioning form an integral part of the service. Every step of the conversation is recorded, and every change in the structure of the posts is recorded, so that it can be played back at any given point.

Teamwork is the very concept Google Wave is based on. Being primarily a communication tool, it quite obviously requires the participation of many people. Whether it is in holding a conversation, or editing an online document, or even playing a game of chess.